

Semi-empirical prediction of dust soiling effects on Agri-PV output power: Particle size effect and implications for adaptive maintenance

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I. INTRODUCTION

Agri-photovoltaic systems combine crop production with renewable energy generation and can support on-farm operations such as irrigation, cooling and postharvest processing. However, dust deposition on the panel surface can reduce photovoltaic performance. In agricultural environments, dust properties depend on soil type and may affect both power loss and cleaning difficulty. Frequent cleaning can also increase water consumption, especially in water-limited regions. Therefore, this study investigates the effect of soil-type-dependent dust deposition on photovoltaic performance and cleaning efficiency under agri-photovoltaic conditions.

II. METHODS

1. Experimental setup

A controlled laboratory experiment was conducted using a photovoltaic module operated with a fixed electrical load. Dust was deposited on the module surface in incremental mass levels ranging from 0 to 5000 gm^{-2} . The PV response under soiled conditions was compared with the clean reference state. Three dust particle-size fractions were investigated: very fine sand, medium sand, and coarse sand. Their median particle diameters are listed in Table 1, while representative images of the deposited dust fractions on the PV module surface are shown in Figure 1.

Table 1: Particle size fractions used in the experiment

Dust fraction	Median particle diameter
Very fine sand	$d_{50} \approx 90 \mu\text{m}$
Medium sand	$d_{50} \approx 125 \mu\text{m}$
Coarse sand	$d_{50} \approx 500 \mu\text{m}$



Fig. 1. Dust particle-size fractions applied to the PV module surface

2. Power ratio metric

The power ratio was used to quantify the relative reduction in power output caused by dust deposition:

$$PR(m) = \frac{P_{soiled}(m)}{P_{clean}}$$

Where $PR(m)$ is the power ratio at dust mass m , $P_{soiled}(m)$ is the output power under soiled conditions and P_{clean} is the output power of the clean module

3. Dust attenuation model

The relationship between dust mass and power ratio was described using a semi-empirical model attenuation function:

$$PR(m) = \exp(-km^b)$$

Where m is the deposited dust mass and k and b are fitted parameters. The parameter k represents the rate of attenuation, while b controls the curvature of the response. Both parameter depend on the dust type and particle size. A larger value of k indicates a faster reduction in output power

References

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III. Results and discussion

The measured and modelled responses showed a clear dependence of photovoltaic (PV) power reduction on particle size. As shown in Fig. 2, very fine sand caused the greatest reduction in output power per unit deposited mass, followed by medium sand, whereas coarse sand produced the slowest decline. This trend indicates that smaller particles caused greater optical obstruction and surface coverage than larger particles. The attenuation response was nonlinear, suggesting that deposited dust mass alone was insufficient to describe PV soiling losses.

Fig. 3 shows the predictive performance of the model for custom particle sizes. By interpolating between calibrated particle-size fractions, the model estimated the response of untested particles and predicted their expected power-loss trends. These results indicate that particle size is an important factor in PV soiling assessment and should be considered when modelling soiling-related power losses in agri-photovoltaic systems.

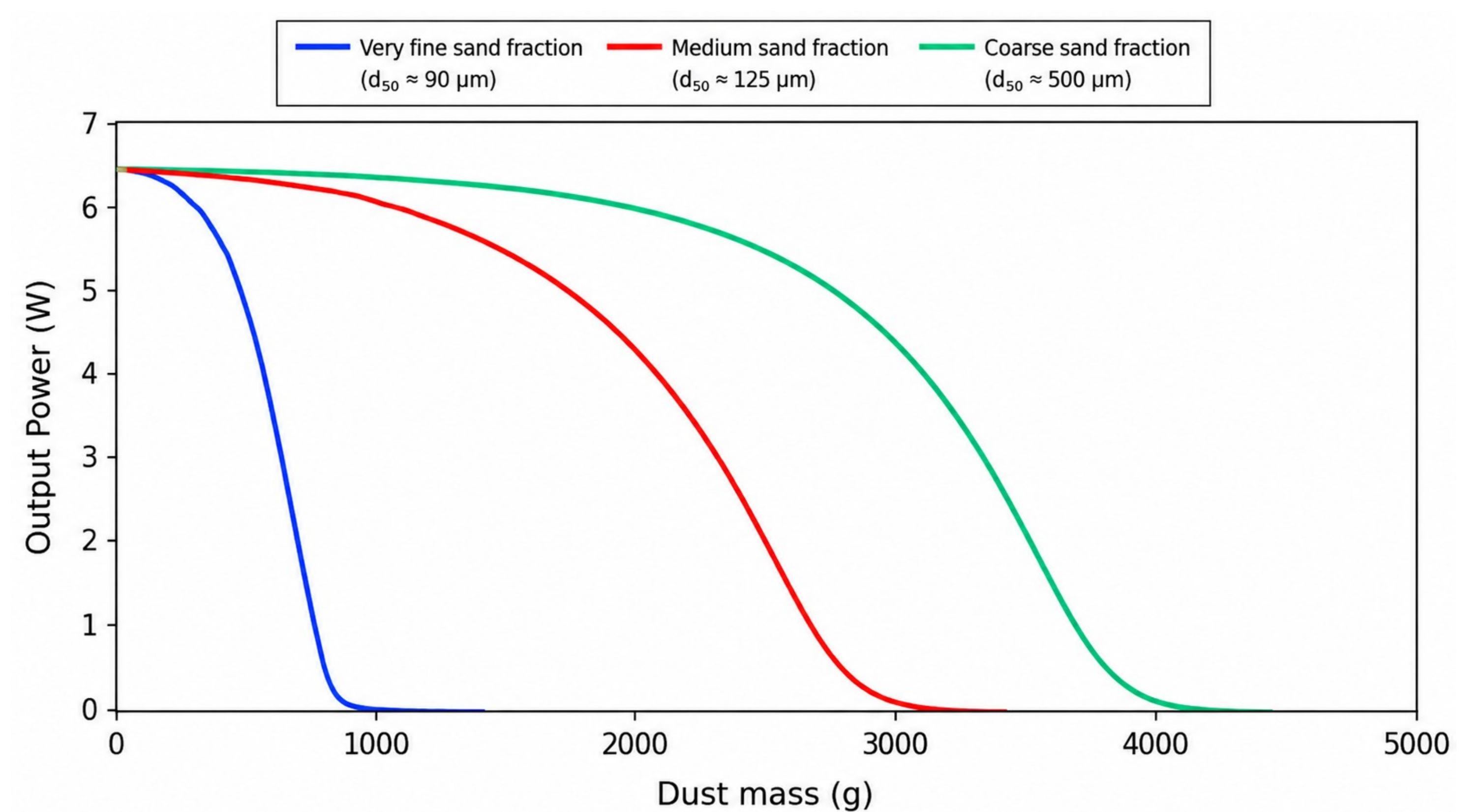


Fig. 2. Model-predicted PV output power as a function of deposited dust mass for the calibrated sand fractions.

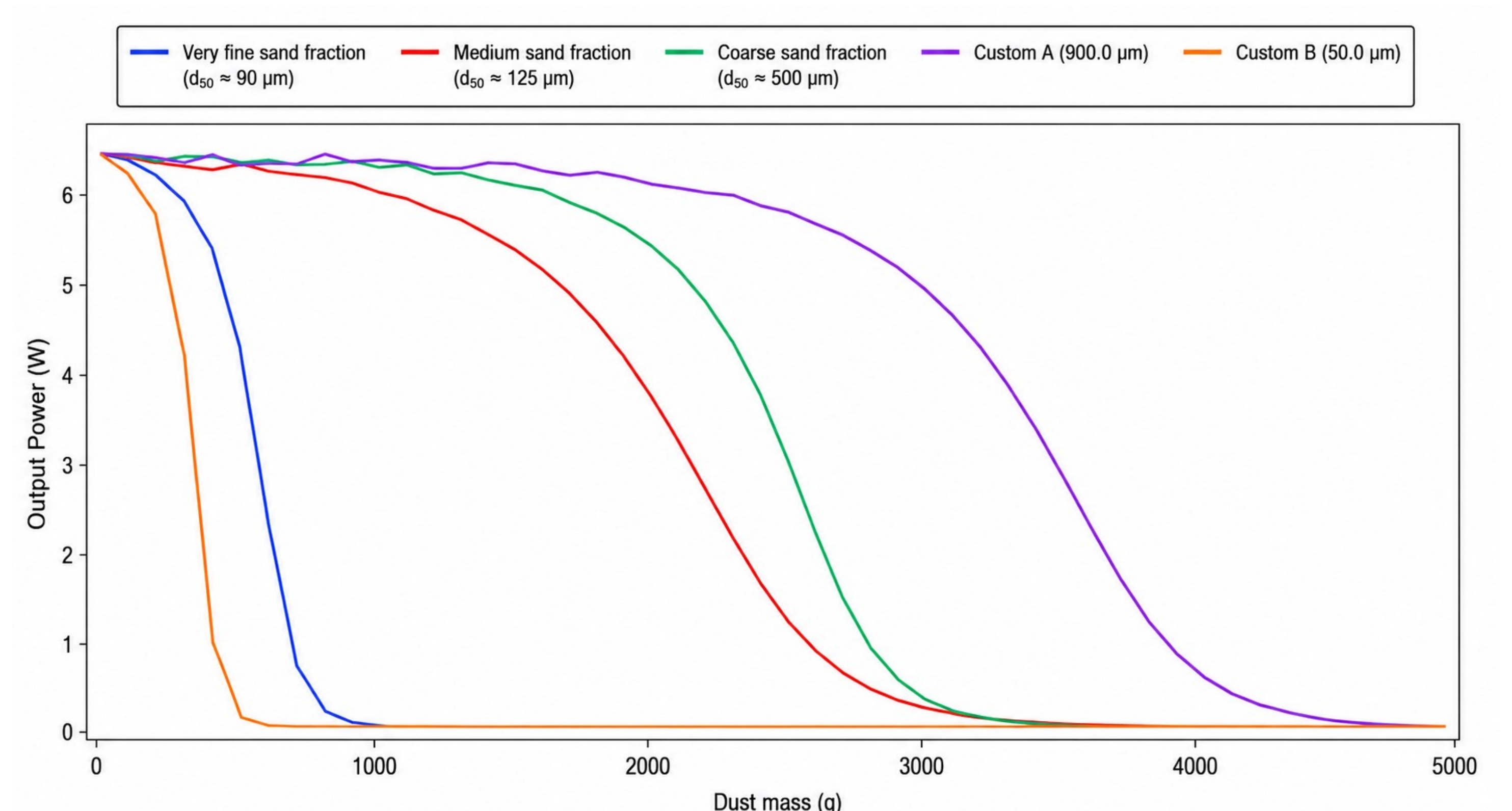


Fig. 3. Model-predicted PV output power including user-defined particle-size scenarios estimated from the semi-empirical model.

IV. Conclusion

PV power loss under dust soiling was strongly influenced by particle size, with fine particles causing the greatest attenuation and coarse particles showing the slowest decline. The nonlinear response indicates that both dust mass and particle size should be considered when estimating soiling losses. The proposed semi-empirical model can support particle-size-dependent power-loss prediction and condition-based cleaning decisions in Agri-PV systems, contributing to more water-efficient and climate-resilient agricultural energy management.



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